

SHARP RANK THRESHOLDS, CRITICAL EXTREMIZERS, AND FIBRES OF EVENTUAL-PERIOD FEEDBACK

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ABSTRACT. For a function on the labels $0, \dots, n-1$, replace its value at each vertex by the length of the eventual cycle reached from that vertex, minus one. Iterating this operator, rather than repeatedly composing a fixed function, always reaches the constant-zero function. We determine the sharp rank needed to have height at least h : the thresholds satisfy $N_2 = 2$ and $N_{h+1} = N_h(N_h + 1)/2$. Every threshold is attained by an explicit permutation, and the resulting bound is sharp at every rank on every larger carrier. At a critical size N_h we classify all deepest states and count them by a factorial recursion. Independently, a target's fibre factors over its level sets into functions with a prescribed cycle length. This gives every empty and nonempty fibre, the first image, and a unique largest fibre at zero of size $(n+1)^{n-1}$. The triangular sequence and all static cycle/forest enumeration are classical and fully subtracted. Complete checks through seven labels and larger critical witnesses provide bounded verification; external status remains `HOLD_EXTERNAL`.

1. THE LABELLED OPERATOR AND ITS SCOPE

Fix $n \geq 1$ and put $\mathcal{X}_n = \{f : \{0, \dots, n-1\} \rightarrow \{0, \dots, n-1\}\}$. For a vertex i , let $\ell_f(i)$ be the length of the unique directed cycle eventually reached by iterating f from i . Loops have length one. Define the simultaneous update

$$(1) \quad (P_n f)(i) = \ell_f(i) - 1.$$

Since $1 \leq \ell_f(i) \leq n$, this is an autonomous self-map of $\mathcal{X}_n$. Numerical labels are retained at every epoch; the output is used as the next function, not as an unlabelled statistic. Write $r(f) = |\text{im } f|$ and denote the constant-zero function by $\mathbf{0}$. The identity function is not $\mathbf{0}$ unless $n = 1$.

Functional-graph decomposition into cycles with rooted trees is classical, as are labelled SET/CYC enumeration and Cayley forest counts [2]. Functional-graph reconstruction by direct products is a neighboring question: Doré et al. use the word “feedback” for a cyclic vertex’s return period but study graph equations up to isomorphism [1]. Equation (1) instead changes the numerical pointers and then recomputes their cycles. The distinction specifies the object; it does not establish priority.

The threshold sequence $2, 3, 6, 21, 231, 26796, \dots$ is already known. Wagner explicitly gives the triangular recurrence in the enumeration of balanced trees and identifies the sequence A007501 [3, Section 4]. The sequence itself, generic rank descent, cycle finding and all static inverse counts receive zero contribution credit. The temporal content here is the sequence’s sharp minimum-rank interpretation, its attained all-rank bound, and the full equality classification at critical sizes. The independent inverse mechanism is the invariant-block decomposition with strict maximal-fibre comparison.

The internal comparison includes P137 rank feedback and P167 least- preimage feedback. A prior word/poset scouting lane excluded canonical statistics written back as states; that lane rule remains unchanged. The present note was separately selected under the central two-axis contract, not by silently waiving that rule. The source record preserves this scope decision and the missing P51–P56 manuscript caveat. No bounded search non-hit or computational success certifies novelty or ownership.

2. CYCLE-SIZE COST AND THE ZERO ATTRACTOR

Let $\mathcal{L}(f)$ be the set of distinct cycle lengths in the functional graph of f , and put $Q(s) = s(s+1)/2$.

Lemma 2.1 (rank packing). *For every $f \in \mathcal{X}_n$,*

$$(2) \quad r(P_n f) = |\mathcal{L}(f)|, \quad r(f) \geq \sum_{d \in \mathcal{L}(f)} d \geq Q(r(P_n f)).$$

Proof. Every cycle has a nonempty basin, so the distinct output values are exactly $d-1$ for $d \in \mathcal{L}(f)$. Choose one cycle for each different length. Their vertices are disjoint and all belong to $\text{im } f$, giving the first inequality. Distinct positive integers with s elements sum to at least $1 + \dots + s = Q(s)$. $\square$

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Proposition 2.2. *Every orbit of P_n reaches $\mathbf{0}$. It is the sole fixed and sole recurrent state, and every eventual period is one.*

Proof. If $r(P_n f) \geq 2$, then $Q(r(P_n f)) > r(P_n f)$, so Lemma 2.1 strictly decreases rank. If the output has rank one, it is constant. A constant function has only a loop as its cycle, and its next output is $\mathbf{0}$. Strict decreases cannot continue indefinitely, and $P_n \mathbf{0} = \mathbf{0}$. No other recurrent state is possible. $\square$

Define $h(f) = \min\{t \geq 0 : P_n^t f = \mathbf{0}\}$, now finite by Proposition 2.2.

Lemma 2.3 (core extension). *Suppose $1 \leq k \leq n$, $g : \{0, \dots, n-1\} \rightarrow \{0, \dots, k-1\}$, and $u = g|_{\{0, \dots, k-1\}}$. For all $t \geq 0$,*

$$(3) \quad (P_n^t g)|_{\{0, \dots, k-1\}} = P_k^t u.$$

For $t \geq 1$, $P_n^t g = \mathbf{0}$ if and only if $P_k^t u = \mathbf{0}$. Thus $h(g) = h(u)$ if $h(u) \geq 1$; if $u = \mathbf{0}$, then $h(g)$ is zero or one according as $g = \mathbf{0}$ or not.

Proof. Every cycle of g lies in the first k labels, and every outside vertex enters that core in one arrow. All periods are at most k , so $P_n g$ again maps into the core and restricts there to $P_k u$. Induction gives (3) and preserves this core-mapping property at every epoch. A period-feedback vector is zero exactly when every cycle of its input is a loop. At every epoch the extension and its core restriction have precisely the same cycles. This proves the equivalence for $t \geq 1$. If $u = \mathbf{0}$, the equivalence says that $P_n g = \mathbf{0}$; time zero must be checked separately, yielding the stated boundary. $\square$

3. SHARP HEIGHT AT EVERY SIZE AND RANK

Define

$$(4) \quad N_2 = 2, \quad N_{h+1} = Q(N_h) \quad (h \geq 2).$$

These are strictly increasing integers.

Theorem 3.1 (sharp rank hierarchy). *For every $h \geq 2$ and every f on any finite carrier,*

$$(5) \quad h(f) \geq h \implies r(f) \geq N_h.$$

There is a height- h permutation on N_h labels. Consequently the maximum height on n labels is

$$(6) \quad H(1) = 0, \quad H(n) = \max\{h \geq 2 : N_h \leq n\} \quad (n \geq 2).$$

More generally, for every $2 \leq r \leq n$, the maximum among rank- r functions on n labels is $H(r)$. At rank one, the maximum is one if $n \geq 2$ and zero if $n = 1$.

Proof. A rank-one function is constant and has height at most one, which proves (5) at $h = 2$. If $h(f) \geq h + 1$, then $h(P_n f) \geq h$, so induction and (2) give $r(f) \geq Q(r(P_n f)) \geq Q(N_h) = N_{h+1}$.

To attain the bound, start with $f_2 = (1, 0)$. Its successive outputs are $(1, 1)$ and $(0, 0)$, so its height is two. Suppose f_h is a height- h permutation on $k = N_h$ labels. For every $j = 0, \dots, k-1$, make a block B_j containing the unique old label $f_h^{-1}(j)$ and exactly j fresh labels. These blocks partition $Q(k) = N_{h+1}$ labels. On each B_j choose any single cycle and let f_{h+1} be their disjoint union. Then $P f_{h+1}$ is constant j on B_j , maps into the first k labels, and restricts there to f_h . Lemma 2.3 shows that $h(P f_{h+1}) = h$, so $h(f_{h+1}) = h + 1$.

For an arbitrary $r \geq 2$, let $h = H(r)$ and take this critical permutation on $k = N_h \leq r$ labels. Extend it to a permutation on r labels by fixing the added labels. Its first period-feedback vector equals $P_k f_h$ on the core and zero outside. It maps into that core. Since $h(P_k f_h) = h - 1 \geq 1$, Lemma 2.3 gives height $h - 1$ for this vector, and height h for the enlarged permutation. Extend further to $n \geq r$ by mapping all new vertices to zero. The new function has rank exactly r and retains height h by the same lemma. This proves the all-rank and all-size attainment statements. Finally, nonzero constant functions exist exactly when $n \geq 2$ and have height one; the only state at $n = 1$ is zero. $\square$

4. ALL DEEPEST STATES AT CRITICAL SIZES

The equality statement in this section concerns $n = N_h$, not arbitrary sizes between two thresholds.

Theorem 4.1 (critical equality and census). *At size N_h , every height- h function is a permutation. For $h \geq 3$ and $k = N_{h-1}$, these are exactly the permutations f satisfying:*

- (1) *there is precisely one cycle of each length $1, 2, \dots, k$;*
- (2) *$u = (P f)|_{\{0, \dots, k-1\}}$ is a height- $(h-1)$ permutation of those k labels.*

If D_h is their number, then

$$(7) \quad D_2 = 1, \quad D_h = D_{h-1}(N_h - N_{h-1})! \quad (h \geq 3).$$

In particular, $D_3 = 1$, $D_4 = 6$, and $D_5 = 6 \cdot 15!$.

Proof. Theorem 3.1 forces $r(f) = N_h$, hence f is a permutation. For $h \geq 3$, $h(Pf) = h - 1$ and the entire chain

$$(8) \quad N_h = r(f) \geq \sum_{d \in \mathcal{L}(f)} d \geq Q(r(Pf)) \geq Q(N_{h-1}) = N_h$$

consists of equalities. The strictly increasing Q forces $r(Pf) = k$. Equality in the distinct-positive-integer sum forces $\mathcal{L}(f) = \{1, \dots, k\}$. Equality in the first inequality says that the selected cycle of each length already exhausts all vertices. There are thus no additional cycles, so condition (1) holds and $\text{im}(Pf) = \{0, \dots, k - 1\}$.

By Lemma 2.3, the restriction u has height $h - 1$. Applying (5) on its $k = N_{h-1}$ labels forces full rank, so it is a permutation and condition (2) holds. Conversely, conditions (1) and (2) make Pf a core extension of a height- $(h - 1)$ function. The same lemma gives $h(Pf) = h - 1$, hence $h(f) = h$. This proves necessity and sufficiency, not just an extremal construction.

Fix an eligible u . Its inverse image of j prescribes one old label in the cycle block B_j of length $j + 1$. Allocate the $N_h - k$ fresh labels among these blocks in $(N_h - k)! / \prod_{j=0}^{k-1} j!$ ways. Each block has $j!$ cyclic orders. Their product cancels the denominator. Different allocations or cyclic orders give different permutations, while the equality characterization shows that every extremizer is obtained. The count per u is therefore $(N_h - k)!$. At $N_2 = 2$ only the transposition has height two, establishing the initial condition and (7). $\square$

5. THE FULL ONE-STEP INVERSE AND ITS MAXIMUM

We isolate the classical enumerative input. Let $R(k, s)$ count forests on a fixed k -element label set directed toward a prescribed set of $s \geq 1$ roots, with exactly one root per component. Then

$$(9) \quad R(k, s) = \begin{cases} 1, & s = k, \\ sk^{k-s-1}, & s < k. \end{cases}$$

A root-preserving Prüfer code proves this formula directly. Repeatedly remove the least nonroot leaf and record its parent until only roots remain. There are $k - s$ recorded labels and the last is a root. Conversely, given any such sequence, remove the least remaining nonroot absent from the remaining sequence and join it to the next recorded parent. At each stage such a nonroot exists: the sequence has as many entries as remaining nonroots and at least one root entry, so it cannot contain all remaining nonroots. A deleted label cannot occur later in the sequence, and edges always lead to a still-present label. The reconstruction therefore yields a forest directed to the specified roots and reverses the encoding. There are sk^{k-s-1} codes when $s < k$. This is standard labelled forest enumeration [2], not a new count.

For $d \geq 1$, let $a_d(k)$ count functions on a prescribed k -element set whose every cycle has length d , with $a_d(0) = 1$. For $k > 0$,

$$(10) \quad a_d(k) = \sum_{c=1}^{\lfloor k/d \rfloor} \frac{k!}{(k - dc)! d^c c!} R(k, dc).$$

Indeed, choose the dc cyclic labels, put c disjoint d -cycles on them, and attach a forest with those prescribed roots. A nonempty such class exists exactly when $k \geq d$: use one d -cycle and attach all remaining vertices to a cyclic vertex. In EGF notation the same classical decomposition reads

$$(11) \quad \sum_{k \geq 0} a_d(k) \frac{z^k}{k!} = \exp\left(\frac{\mathcal{T}(z)^d}{d}\right), \quad \mathcal{T}(z) = z \exp(\mathcal{T}(z)).$$

Theorem 5.1 (every target fibre and image). *For $g \in \mathcal{X}_n$, write $B_j = g^{-1}(j)$ and $k_j = |B_j|$. Then*

$$(12) \quad |P_n^{-1}(g)| = \prod_{j=0}^{n-1} a_{j+1}(k_j).$$

The target lies in $\text{im } P_n$ exactly when each nonempty B_j satisfies $k_j \geq j + 1$. Consequently

$$(13) \quad |\text{im } P_n| = \sum_{\substack{\sum_{j=0}^{n-1} k_j = n \\ k_j = 0 \text{ or } k_j \geq j+1}} \frac{n!}{\prod_{j=0}^{n-1} k_j!}.$$

Proof. An arrow $i \mapsto f(i)$ does not change the eventual cycle reached. Thus $P_n f = g$ implies that f preserves every B_j and all cycles of $f|_{B_j}$ have length $j + 1$. Conversely these conditions force the period of every vertex in B_j to be $j + 1$, so they imply $P_n f = g$. The restrictions on distinct labelled blocks are independent, proving (12). The nonemptiness criterion for $a_d(k)$ gives the image condition. Each admissible vector of block sizes has $n! / \prod_j k_j!$ target words, proving (13). $\square$

Theorem 5.2 (strict fibre extremality). *For every $n \geq 1$, the unique largest fibre is at $g = \mathbf{0}$, with size*

$$(14) \quad |P_n^{-1}(\mathbf{0})| = (n + 1)^{n-1}.$$

Proof. Let $c_d(k)$ count connected functions on k labels with sole cycle length d . For $k \geq d$, selecting the cyclic vertices and attaching their rooted forest gives

$$(15) \quad c_d(k) = (k)_d k^{k-d-1}, \quad c_1(k) = k^{k-1},$$

where $(k)_d = k(k-1) \cdots (k-d+1)$ and the first expression at $k = d$ is $(d-1)!$. Their ratio is $(k)_d / k^d \leq 1$, strictly less than one when $d \geq 2$. For $k < d$, $c_d(k) = 0$. Hence the EGF of connected d -cycle components is coefficientwise bounded by the rooted-tree EGF. Taking SET exponentials preserves this inequality: all coefficients are nonnegative and every product term is bounded term by term. Moreover, the coefficient at every $k \geq d \geq 2$ is strictly smaller, already in the single-component term. Thus

$$(16) \quad a_d(k) \leq a_1(k),$$

with strict inequality for every nonempty admissible block with $d \geq 2$.

A product of rooted-forest counts on two or more prescribed nonempty blocks is strictly smaller than the rooted-forest count on their union. Disjoint union is an injection, while a forest with one edge between two different blocks is excluded. Applying this fact and (16) to (12), equality with the full rooted-forest count requires a single block and $d = 1$. This is exactly $g = \mathbf{0}$. Its fibre is the set of functions having only loops as cycles, equivalently rooted forests with a loop at each root. Adjoining one extra vertex joined to every root gives a bijection with trees on $n + 1$ labels rooted at that extra vertex. Cayley's formula gives $(n + 1)^{n-1}$. At $n = 1$ this is one and the zero target is the only target. Unsupported targets have fibre zero and cannot be maximizers. $\square$

6. EXACT CONTROLS AND CLAIM CEILINGS

TABLE 1. Complete endofunction boxes for the paper-local verifier.

n	States	First image	Maximum height	Maximum fibre
1	1	1	0	1
2	4	2	2	3
3	27	6	3	16
4	256	18	3	125
5	3125	60	3	1296
6	46656	240	4	16807
7	823543	1085	4	262144

The standard-library verifier computes the update by peeling noncyclic vertices and then propagating cycle periods backwards. Per-vertex orbit tracing supplies a second implementation on all inputs through six labels and a stated prefix of the seven-label box. The full source boxes check rank packing, all-rank maxima, actual target fibres and the image. Every target, including absent targets, is checked through six labels. Critical-size height counts are checked at sizes two, three and six. Recursive witnesses on 2, 3, 6, 21, 231, 26796 labels attain heights two through seven, with further leaf extensions. Two fresh processes are compared with the paper-local canonical transcript. All-size validity rests on the proofs, not on extrapolation from these boxes.

No scalar rank evolution law is asserted: (2) is an inequality. Nor is there a closed formula for each individual height or an all-time target inverse atlas. The equality classification is complete only at the

critical sizes N_h . For instance, the two functions $(0, 1, 1)$ and $(1, 0, 1)$ have the same value histogram, but their period feedback vectors are $(0, 0, 0)$ and $(1, 1, 1)$. This refutes a histogram factor only; it does not waive the earlier lane’s statistic-writeback filter or establish novelty. Static formulas (9)–(11) are classical inputs, not separate discoveries. External status remains `OWNER_AMBER` / `HOLD_EXTERNAL`.

Reproducibility and disclosure. The anonymous source, verifier, fixed transcript, source-scope audit and deterministic build instructions are provided with this internal note. No empirical or human-subject data are used. AI-assisted proof exploration, drafting and exact verification were used; author checks are not human expert review. Authorship, funding and conflicts are not inferred for this internal draft. The known triangular sequence and classical functional-graph counts remain attributed throughout.

REFERENCES

- [1] François Doré, Enrico Formenti, Antonio E. Porreca, and Sara Riva. Algorithmic reconstruction of discrete dynamics. arXiv:2208.08310v2, 2022. URL <https://arxiv.org/html/2208.08310v2>. Version 2, 6 September 2022.
- [2] Philippe Flajolet and Robert Sedgewick. *Analytic Combinatorics*. Cambridge University Press, 2009. doi: 10.1017/CBO9780511801655. URL <https://ac.cs.princeton.edu/home/>.
- [3] Stephan Wagner. Enumeration of highly balanced trees. *Ars Combinatoria*, 114:15–32, 2014. URL <https://math.sun.ac.za/swagner/balanced.pdf>.